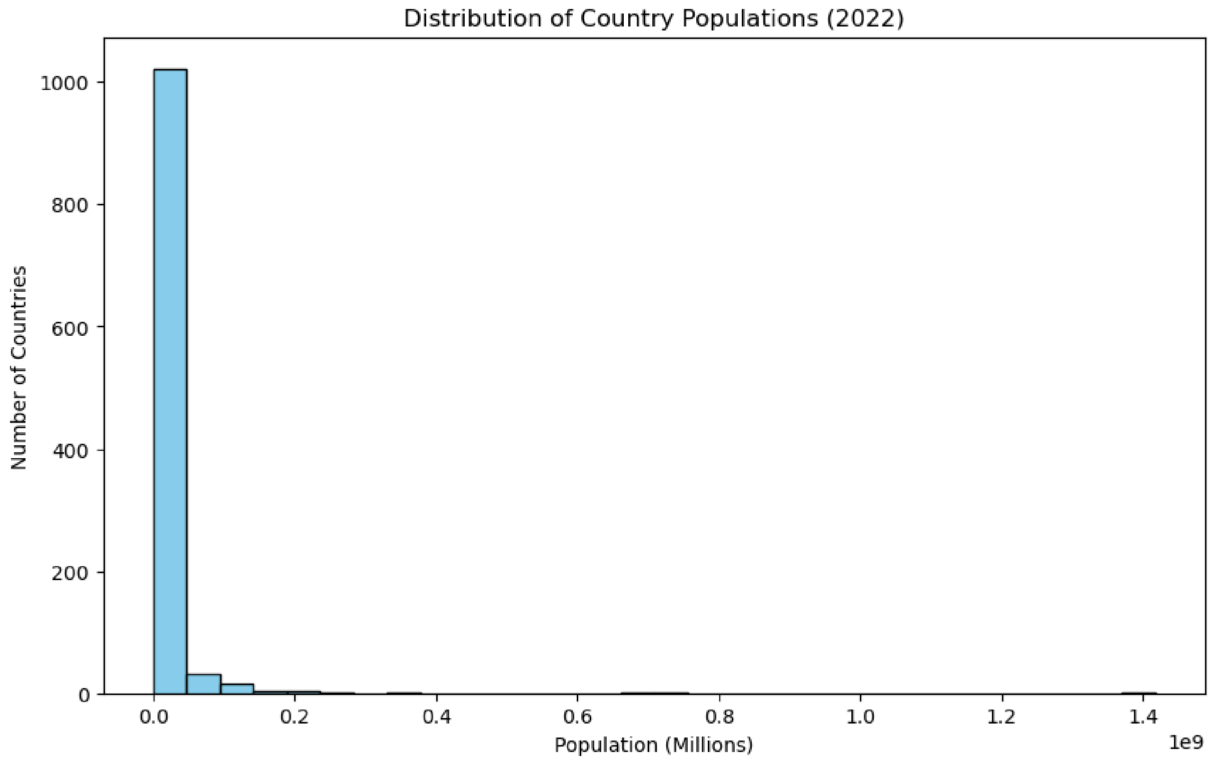
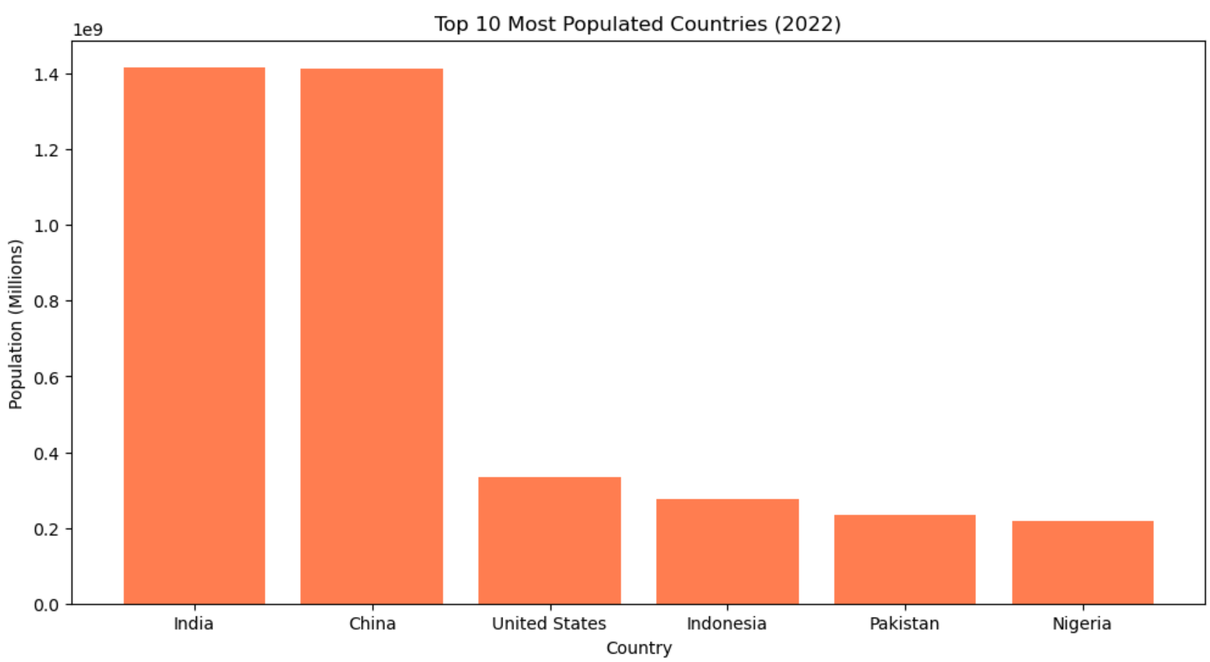
In [9]:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv(r'C:\Users\LOQ\Downloads\worldpopulationdata.csv')

year_to_plot = '2022'
country_col = 'Country Name'

# Clean the data: Remove any rows where the population data for our chosen year is
df_clean = df.dropna(subset=[year_to_plot])

# CHART 1: HISTOGRAM (Continuous Variable)
# Showing the distribution of country populations

plt.figure(figsize=(10, 6))
plt.hist(df_clean[year_to_plot], bins=30, color='skyblue', edgecolor='black')

plt.title(f'Distribution of Country Populations ({year_to_plot})')
plt.xlabel('Population (Millions)')
plt.ylabel('Number of Countries')

#first chart
plt.show()

# CHART 2: BAR CHART (Categorical Variable)

# Showing the Top 10 most populated countries

# setting it in decending format from highest to lowest data.
top_10_countries = df_clean.sort_values(by=year_to_plot, ascending=False).head(10)

plt.figure(figsize=(12, 6))
plt.bar(top_10_countries[country_col], top_10_countries[year_to_plot], color='coral

plt.title(f'Top 10 Most Populated Countries ({year_to_plot})')
plt.xlabel('Country')
plt.ylabel('Population (Millions)')

# NOTE: If this country name given on X axix does overlap the use below code to rot
# Rotate the country names by 45 degrees so they don't overlap
# plt.xticks(rotation=45)

# Display the second chart
plt.show()
```

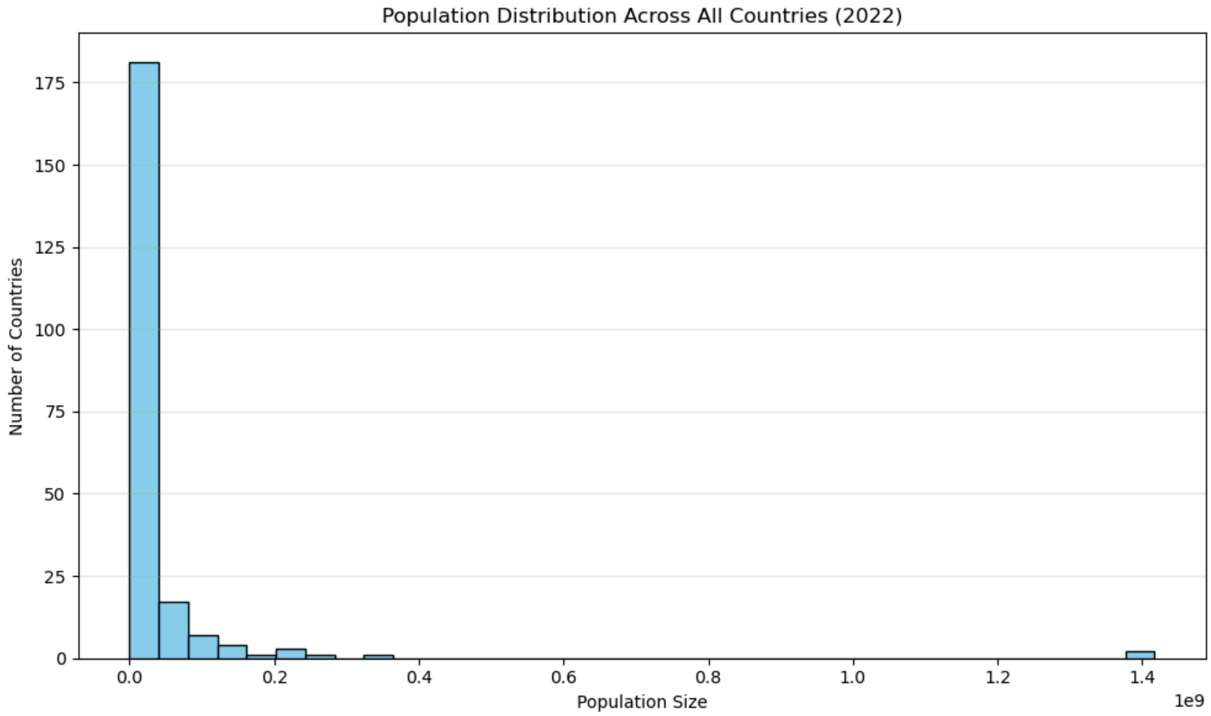




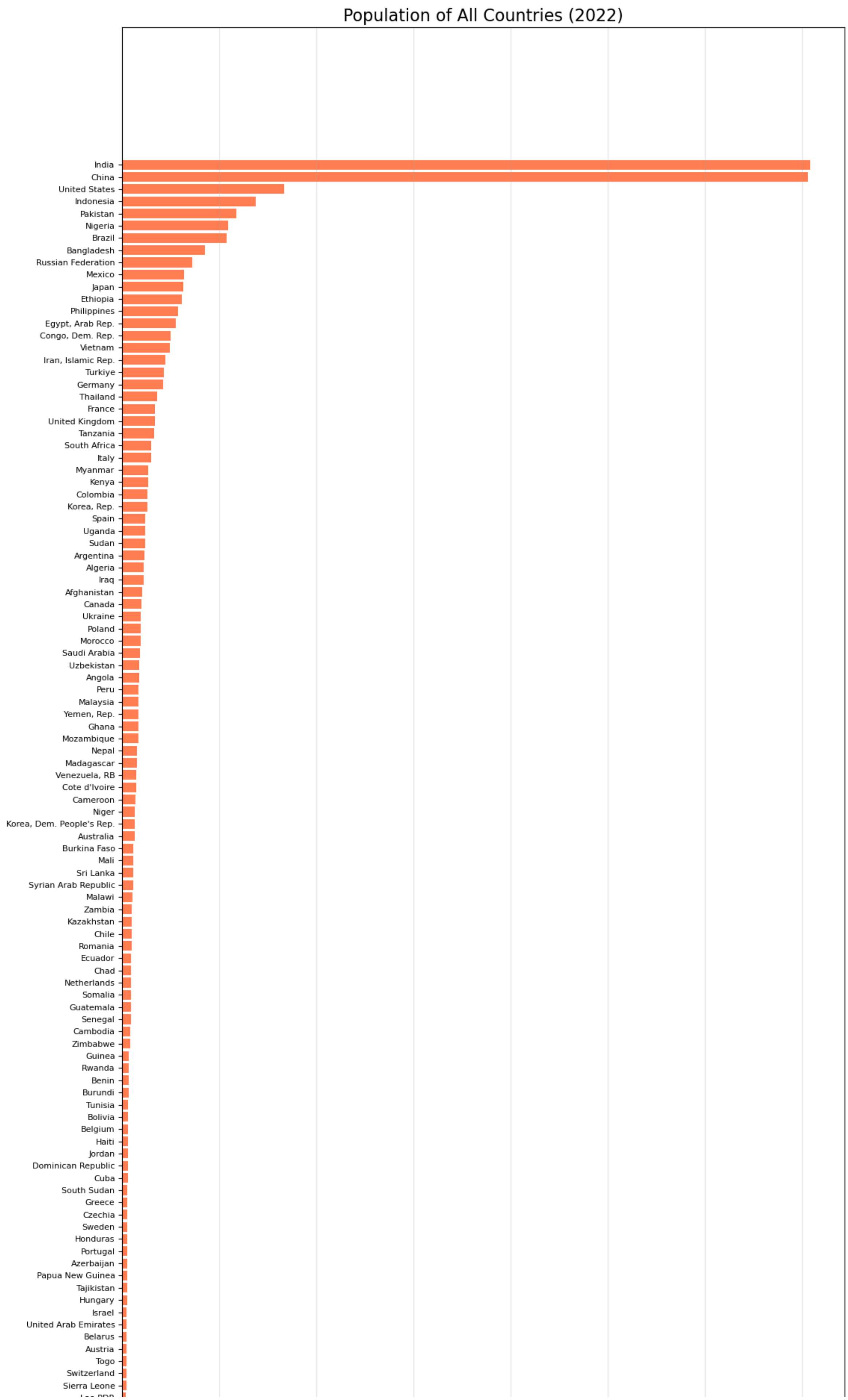
In [6]:

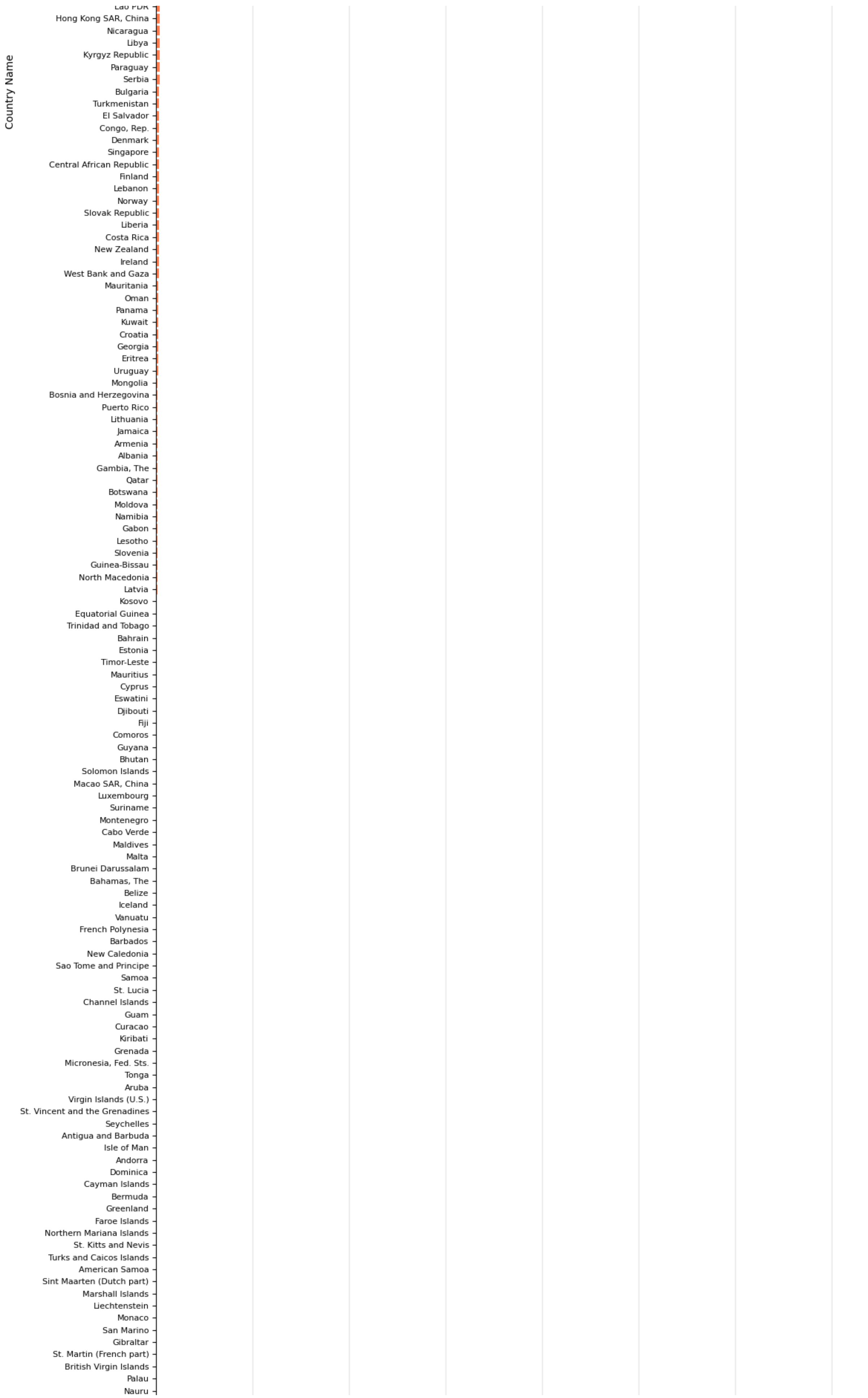
```
# 2. CRITICAL FILTER: Keep ONLY the total population rows
# This prevents mixing population counts with percentage rows (like male/female %)
df_population = df[df['Series Code'] == 'SP.POP.TOTL'].copy()

year_to_plot = '2022'
country_col = 'Country Name'

# 3. Clean the data
# Convert the population column to numbers and drop missing rows
df_population[year_to_plot] = pd.to_numeric(df_population[year_to_plot], errors='co
df_clean = df_population.dropna(subset=[year_to_plot]).copy()
```

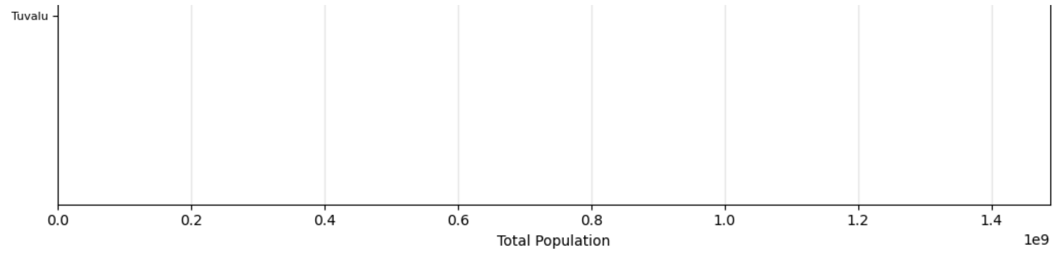
```
# Sort countries from smallest to largest population so the bar chart looks organiz
df_clean = df_clean.sort_values(by=year_to_plot, ascending=True)

# ==========================================
# CHART 1: HISTOGRAM (All Countries)
# ==========================================
plt.figure(figsize=(10, 6))

# Plots how many countries fall into specific population size buckets
plt.hist(df_clean[year_to_plot], bins=35, color='skyblue', edgecolor='black')

plt.title(f'Population Distribution Across All Countries ({year_to_plot})')
plt.xlabel('Population Size (Millions)')
plt.ylabel('Number of Countries')
plt.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()


# ==========================================
# CHART 2: BAR CHART (All Countries)
# ==========================================
# We make the figure very tall (height=40) so all 200+ country labels fit comfortab
plt.figure(figsize=(12, 40))

# 'barh' creates horizontal bars: Y-axis is country names, X-axis is population
plt.barh(df_clean[country_col], df_clean[year_to_plot], color='coral')

plt.title(f'Population of All Countries ({year_to_plot})', fontsize=16)
plt.xlabel('Total Population (Millions)')
plt.ylabel('Country Name')

# Shrink the text size slightly so names do not overlap on the side
plt.yticks(fontsize=8)
plt.grid(axis='x', alpha=0.3)

plt.tight_layout()
plt.show()
```



Population Distribution Across All Countries (2022)
Number of Countries
0
25
50
75
100
125
150
175
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1e9
Population Size


Population of All Countries (2022)
India
China
United States
Indonesia
Pakistan
Nigeria
Brazil
Bangladesh
Russian Federation
Mexico
Japan
Ethiopia
Philippines
Egypt, Arab Rep.
Congo, Dem. Rep.
Vietnam
Iran, Islamic Rep.
Turkiye
Germany
Thailand
France
United Kingdom
Tanzania
South Africa
Italy
Myanmar
Kenya
Colombia
Korea, Rep.
Spain
Uganda
Sudan
Argentina
Algeria
Iraq
Afghanistan
Canada
Ukraine
Poland
Morocco
Saudi Arabia
Uzbekistan
Angola
Peru
Malaysia
Yemen, Rep.
Ghana
Mozambique
Nepal
Madagascar
Venezuela, RB
Cote d'Ivoire
Cameroon
Niger
Korea, Dem. People's Rep.
Australia
Burkina Faso
Mali
Sri Lanka
Syrian Arab Republic
Malawi
Zambia
Kazakhstan
Chile
Romania
Ecuador
Chad
Netherlands
Somalia
Guatemala
Senegal
Cambodia
Zimbabwe
Guinea
Rwanda
Benin
Burundi
Tunisia
Bolivia
Belgium
Haiti
Jordan
Dominican Republic
Cuba
South Sudan
Greece
Czechia
Sweden
Honduras
Portugal
Azerbaijan
Papua New Guinea
Tajikistan
Hungary
Israel
United Arab Emirates
Belarus
Austria
Togo
Switzerland
Sierra Leone


Country Name
Hong Kong SAR, China
Nicaragua
Libya
Kyrgyz Republic
Paraguay
Serbia
Bulgaria
Turkmenistan
El Salvador
Congo, Rep.
Denmark
Singapore
Central African Republic
Finland
Lebanon
Norway
Slovak Republic
Liberia
Costa Rica
New Zealand
Ireland
West Bank and Gaza
Mauritania
Oman
Panama
Kuwait
Croatia
Georgia
Eritrea
Uruguay
Mongolia
Bosnia and Herzegovina
Puerto Rico
Lithuania
Jamaica
Armenia
Albania
Gambia, The
Qatar
Botswana
Moldova
Namibia
Gabon
Lesotho
Slovenia
Guinea-Bissau
North Macedonia
Latvia
Kosovo
Equatorial Guinea
Trinidad and Tobago
Bahrain
Estonia
Timor-Leste
Mauritius
Cyprus
Eswatini
Djibouti
Fiji
Comoros
Guyana
Bhutan
Solomon Islands
Macao SAR, China
Luxembourg
Suriname
Montenegro
Cabo Verde
Maldives
Malta
Brunei Darussalam
Bahamas, The
Belize
Iceland
Vanuatu
French Polynesia
Barbados
New Caledonia
Sao Tome and Principe
Samoa
St. Lucia
Channel Islands
Guam
Curacao
Kiribati
Grenada
Micronesia, Fed. Sts.
Tonga
Aruba
Virgin Islands (U.S.)
St. Vincent and the Grenadines
Seychelles
Antigua and Barbuda
Isle of Man
Andorra
Dominica
Cayman Islands
Bermuda
Greenland
Faroe Islands
Northern Mariana Islands
St. Kitts and Nevis
Turks and Caicos Islands
American Samoa
Sint Maarten (Dutch part)
Marshall Islands
Liechtenstein
Monaco
San Marino
Gibraltar
St. Martin (French part)
British Virgin Islands
Palau
Nauru



In [ ]: